Fusion Categories and SymTFT: Homework #6

Non-invertible symmetries in the Ising model and anyon chains
March 8, 2026

Problem 1: Deformation of $1+1$d Ising chain

Consider a 1D transverse-field Ising chain of length L with Pauli operators X_j, Z_j and periodic boundary conditions. In the presence of a λ deformation, the Hamiltonian is

$$H(g,\lambda) = -g\sum_{j=1}^{L} X_j - \sum_{j=1}^{L} Z_j Z_{j+1} + \frac{\lambda}{2}\sum_{j=1}^{L}\left(X_{j-1}Z_jZ_{j+1} + Z_{j-1}Z_jX_{j+1}\right). \tag{1}$$

The deformation preserves both the global $\mathbb{Z}_2$ symmetry and a non-invertible symmetry D (the latter is only a symmetry of the Hamiltonian at $g = g_c = 1$). This deformation was first studied by O'Brien and Fendley [ArXiv: 1712.06662] and is irrelevant for small λ, leaving the CFT intact. For sufficiently large λ ($\lambda > \lambda_c \approx 0.43$), however, the system becomes gapped (see figure below).

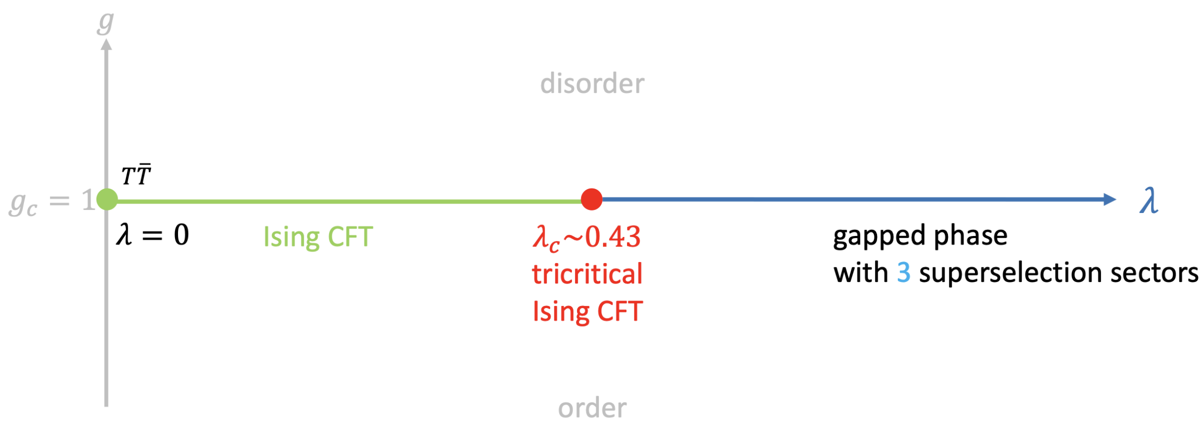

Figure 1: Phase diagram of Ising model with parameters g and λ. Figure taken from Shu-Heng Shao's lecture notes.

In the following, you will show that for sufficiently large deformation parameter (specifically, $\lambda = 1$), the Hamiltonian $H(g = 1, \lambda = 1)$ has three gapped ground states given by

$$|++\cdots+\rangle, \quad |00\ldots0\rangle, \quad |11\ldots1\rangle. \tag{2}$$

(a) First, define the following local operators:

$$P_j^{(1)} = \frac{1}{2}(1 - Z_jZ_{j+1})(1 - X_{j-1}), \quad P_j^{(2)} = \frac{1}{2}(1 - Z_{j-1}Z_j)(1 - X_{j+1}). \tag{3}$$

Show that at $\lambda = 1$, the Hamiltonian can be written (up to an additive constant) as

$$H(g = 1, \lambda = 1) = \sum_{j=1}^{L}\left(P_j^{(1)} + P_j^{(2)}\right). \tag{4}$$

(b) Show that $P_j^{(1)}$ and $P_j^{(2)}$ are projectors, i.e., they are Hermitian and satisfy $P^2 = P$.

(c) Using the fact that projectors have eigenvalues 0 or 1, argue that the ground state energy of $H(\lambda = 1)$ is lower bounded by zero.

(d) Show explicitly that the following three product states

$$|++\cdots+\rangle, \quad |00\ldots0\rangle, \quad |11\ldots1\rangle \tag{5}$$

are annihilated by all $P_j^{(1)}$ and $P_j^{(2)}$, and therefore form exactly degenerate ground states of the Hamiltonian.

This corresponds to a gapped phase that spontaneously break the non-invertible symmetry D (as well as $\mathbb{Z}_2$).

Problem 2: Ising anyon chain

[Adapted from Colleen Delaney's lecture notes]

(a) Consider the anyon chain build on the Ising unitary *fusion* category with reference simple non-invertible object σ. Show that, under periodic boundary conditions the total Hilbert space $\mathcal{H}_{total}$ is only nontrivial for anyon chains with an even number of edges.

(b) Without computing formulas for how the symmetry generators D^1, D^σ, and D^ψ act on $\mathcal{H}_{total}$, show that $D^\sigma D^\sigma = D^1 + D^\psi$. *Hint:* how can you use the graphical calculus in a skeletal fusion category to resolve the fusion of two loops into the anyon chain into a fusion of a single loop?